

- 6 Consider the multiple regression model containing three independent variables, under Assumptions MLR.1 through MLR.4:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u.$$

You are interested in estimating the sum of the parameters on x_1 and x_2 ; call this $\theta_1 = \beta_1 + \beta_2$.

- (i) Show that $\hat{\theta}_1 = \hat{\beta}_1 + \hat{\beta}_2$ is an unbiased estimator of θ_1 .
- (ii) Find $\text{Var}(\hat{\theta}_1)$ in terms of $\text{Var}(\hat{\beta}_1)$, $\text{Var}(\hat{\beta}_2)$, and $\text{Corr}(\hat{\beta}_1, \hat{\beta}_2)$.

$$\begin{aligned} (1) \quad E(\hat{\theta}_1) &= E(\hat{\beta}_1 + \hat{\beta}_2) = E(\hat{\beta}_1) + E(\hat{\beta}_2) \\ &= \beta_1 + \beta_2 = \theta_1 \\ (2) \quad \text{Var}(\hat{\theta}_1) &= \text{Var}(\hat{\beta}_1 + \hat{\beta}_2) \\ &= \text{Var}(\hat{\beta}_1) + \text{Var}(\hat{\beta}_2) + 2\text{Cov}(\hat{\beta}_1, \hat{\beta}_2) \\ &= \text{Var}(\hat{\beta}_1) + \text{Var}(\hat{\beta}_2) + 2\text{Corr}(\hat{\beta}_1, \hat{\beta}_2) \sqrt{\text{Var}(\hat{\beta}_1) \text{Var}(\hat{\beta}_2)} \end{aligned}$$

- 10 Suppose that you are interested in estimating the ceteris paribus relationship between y and x_1 . For this purpose, you can collect data on two control variables, x_2 and x_3 . (For concreteness, you might think of y as final exam score, x_1 as class attendance, x_2 as GPA up through

the previous semester, and x_3 as SAT or ACT score.) Let $\tilde{\beta}_1$ be the simple regression estimate from y on x_1 and let $\hat{\beta}_1$ be the multiple regression estimate from y on x_1, x_2, x_3 .

- (i) If x_1 is highly correlated with x_2 and x_3 in the sample, and x_2 and x_3 have large partial effects on y , would you expect $\tilde{\beta}_1$ and $\hat{\beta}_1$ to be similar or very different? Explain.
- (ii) If x_1 is almost uncorrelated with x_2 and x_3 , but x_2 and x_3 are highly correlated, will $\tilde{\beta}_1$ and $\hat{\beta}_1$ tend to be similar or very different? Explain.
- (iii) If x_1 is highly correlated with x_2 and x_3 , and x_2 and x_3 have small partial effects on y , would you expect $\text{se}(\tilde{\beta}_1)$ or $\text{se}(\hat{\beta}_1)$ to be smaller? Explain.
- (iv) If x_1 is almost uncorrelated with x_2 and x_3 , x_2 and x_3 have large partial effects on y , and x_2 and x_3 are highly correlated, would you expect $\text{se}(\tilde{\beta}_1)$ or $\text{se}(\hat{\beta}_1)$ to be smaller? Explain.

$$(1) \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3$$

$$\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$$

$$\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \tilde{\delta}_1 + \hat{\beta}_3 \tilde{\delta}_1$$

$$\tilde{x}_2 = \tilde{\delta}_0 + \tilde{\delta}_1 x_1$$

$$\tilde{x}_3 = \tilde{\delta}_0 + \tilde{\delta}_1 x_1$$

x_1 is highly correlated with x_2, x_3

$\tilde{\delta}_1, \tilde{\delta}_1$ are large 第 14 页 $\hat{\beta}_2 \tilde{\delta}_1 + \hat{\beta}_3 \tilde{\delta}_1$ are large

So very different

(2) x_1 is almost uncorrelated with x_2 and x_3

$\tilde{\delta}_1, \tilde{\delta}_1$ are small

$\hat{\beta}_2 \tilde{\delta}_1 + \hat{\beta}_3 \tilde{\delta}_1$ are small

similar

$$(3) \text{Var}(\tilde{\beta}_1) = \frac{\sigma^2}{SST_1}$$

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{SST_1(1-R_1^2)}$$

$se(\tilde{\beta}_1)$ is smaller

x_1 is highly correlated with x_2 and x_3

R^2 is close to 1 第 13 页 x_2 and x_3 have small partial effects on y , σ^2 in the two equations are similar. $\text{Var}(\hat{\beta}_1)$ is larger.

(4) x_1 is almost uncorrelated with x_2 and x_3

R^2 is close to 0

x_2 and x_3 have large partial effects on y

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{SST_1(1-R_1^2)} \quad \sigma^2 \text{ is smaller}$$

$se(\hat{\beta}_1)$ is smaller.

- 8 Consider the multiple regression model with three independent variables, under the classical linear model assumptions MLR.1 through MLR.6:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u.$$

You would like to test the null hypothesis $H_0: \beta_1 - 3\beta_2 = 1$.

- Let $\hat{\beta}_1$ and $\hat{\beta}_2$ denote the OLS estimators of β_1 and β_2 . Find $\text{Var}(\hat{\beta}_1 - 3\hat{\beta}_2)$ in terms of the variances of $\hat{\beta}_1$ and $\hat{\beta}_2$ and the covariance between them. What is the standard error of $\hat{\beta}_1 - 3\hat{\beta}_2$?
- Write the t statistic for testing $H_0: \beta_1 - 3\beta_2 = 1$.
- Define $\theta_1 = \beta_1 - 3\beta_2$ and $\hat{\theta}_1 = \hat{\beta}_1 - 3\hat{\beta}_2$. Write a regression equation involving β_0 , θ_1 , β_2 , and β_3 that allows you to directly obtain $\hat{\theta}_1$ and its standard error.

$$(1) \text{Var}(\hat{\beta}_1 - 3\hat{\beta}_2) = \text{Var}(\hat{\beta}_1) + 9\text{Var}(\hat{\beta}_2) - 6\text{Cov}(\hat{\beta}_1, \hat{\beta}_2)$$

$$\text{se}(\hat{\beta}_1 - 3\hat{\beta}_2) = \sqrt{\text{Var}(\hat{\beta}_1) + 9\text{Var}(\hat{\beta}_2) - 6\text{Cov}(\hat{\beta}_1, \hat{\beta}_2)}$$

$$(2) t = \frac{\hat{\beta}_1 - 3\hat{\beta}_2 - 1}{\text{se}(\hat{\beta}_1 - 3\hat{\beta}_2)} = \frac{\hat{\beta}_1 - 3\hat{\beta}_2 - 1}{\sqrt{\text{Var}(\hat{\beta}_1) + 9\text{Var}(\hat{\beta}_2) - 6\text{Cov}(\hat{\beta}_1, \hat{\beta}_2)}}$$

$$(3) \begin{aligned} \theta_1 &= \beta_1 - 3\beta_2 & y &= \beta_0 + (\theta_1 + 3\beta_2)x_1 + \beta_2 x_2 + \beta_3 x_3 + u \\ \beta_1 &= \theta_1 + 3\beta_2 & &= \beta_0 + \theta_1 x_1 + 3\beta_2 x_1 + \beta_2 x_2 + \beta_3 x_3 + u \\ t &= \frac{\hat{\theta}_1 - 1}{\text{se}(\hat{\theta}_1)} & \leftarrow &= \beta_0 + \theta_1 x_1 + \beta_2 (3x_1 + x_2) + \beta_3 x_3 + u \end{aligned}$$